

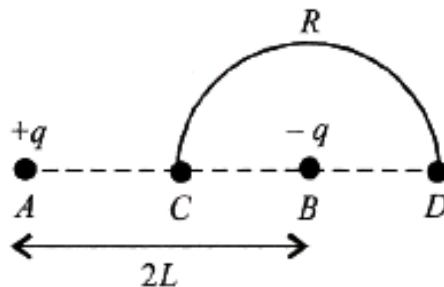
FREEDOM INTERNATIONAL SCHOOL

WORKSHEET

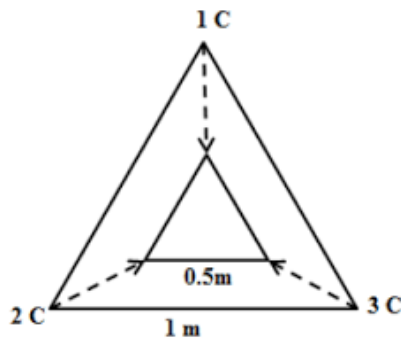
PHYSICS

ELECTROSTATIC POTENTIAL AND CAPACITANCE

1. Electric field intensity at point 'B' due to a point charge 'Q' kept at point 'A' is 24 N/C and the electric potential at point 'B' due to same charge is 12 J/C . Calculate the distance AB and also the magnitude of charge Q.
2. Calculate the electric potential at the centre of a square of side $\sqrt{2} \text{ m}$, having charges $100 \mu\text{C}$, $-50 \mu\text{C}$, $20 \mu\text{C}$ and $-60 \mu\text{C}$ at the four corners of the square.
3. A charge of $24 \mu\text{C}$ is given to a hollow metallic sphere of radius 0.2 m . Find the potential (i) at the surface of the sphere, and (ii) at a distance of 0.1 cm from the centre of the sphere.
4. An electric dipole of length 2 cm is placed with its axis making an angle of 60° to a uniform electric field of 10^5 N/C . If it experiences a torque of $8\sqrt{3} \text{ Nm}$, calculate
 - (i) the magnitude of the charge on the dipole, and
 - (ii) the potential energy of the dipole.
5. Find the amount of work done in rotating an electric dipole of dipole moment $3.2 \times 10^{-8} \text{ Cm}$, from its position of stable equilibrium, to the position of unstable equilibrium, in a uniform electric field of intensity 10^4 N/C .
6. Two charges $3 \times 10^{-8} \text{ C}$ and $-2 \times 10^{-8} \text{ C}$ are located 15 cm apart. At what point on the line joining the two charges is the electric potential zero?
7. Charges q and $-q$ are placed at points A and B respectively which are at a distance $2L$ apart and C is the mid-point between A and B. The work done in moving a charge $+Q$ along the semicircle CRD with B as centre is _____.



8. Three-point charges of 1 C , 2 C and 3 C are placed at the corners of an equilateral triangle of side 1 m . Calculate the work required to move these charges to the corners of a smaller equilateral triangle of sides 0.5 m as shown in figure.



9. Two-point charges of $+3.2 \times 10^{-19} \text{ C}$ and $-3.2 \times 10^{-19} \text{ C}$ are placed at a distance of $2.4 \times 10^{-10} \text{ m}$ from each other, forming an electric dipole. Find (i) the electric dipole moment (ii) potential energy of the dipole in equilibrium position (iii) work done in turning the dipole through 180° from the equilibrium position.
10. The electric field in a region is given by $E = \frac{A}{x^3} \hat{i}$. Write the SI unit for A. Write an expression for the potential in the region assuming the potential at infinity to be zero.